

Floatation.

- Fluids - It is a substance which flows under the action of Applied force. (Liquid + Gas).
- Pressure - Normal force acting per unit area on the surface of body.

$$P = \frac{F}{A} \rightarrow \frac{\text{Thrust}}{A}$$

- ⇒ ~~Thrust~~ Thrust = Normal force on the surface of body.

- ⇒ Unit of Pressure - $P = \frac{F}{A} = \frac{N}{m^2} \Rightarrow \boxed{\text{Pascal}}$

- ⇒ Density - ρ (Rho)

$\rho = \frac{m}{V}$ - density is defined as mass per unit volume.

- Unit - kg/m^3

- *⇒ Density of Water = $1000 kg/m^3$ at $4^\circ C$.

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★ → ~~R.D.~~ Relative density = $\frac{\text{Density of the substance}}{\text{Density of water}}$

- Density of substance = R.D. × Density of water.

Q) R.D. of gold is 19.6

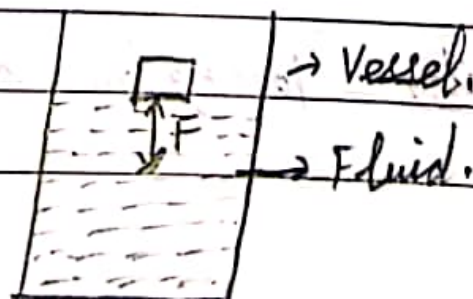
$$\begin{aligned}\therefore \text{Density of gold} &= \text{R.D.} \times \text{Density of substance} \\ &= 19.6 \times 1000 \text{ kg/m}^3 \\ &= 19600 \text{ kg/m}^3.\end{aligned}$$

⇒ Gold is 19.6 times heavier than equal volume of Water.

Q. R.D. of silver (Ag) is 10.3.

$$\begin{aligned}\text{Density of silver} &: \text{R.D.} \times \text{D. of Water} \\ &= 10.3 \times 1000 \text{ kg m}^{-3} \\ &= 10300 \text{ kg m}^{-3}\end{aligned}$$

⇒ Buoyancy and Buoyant force:



- The upward force exerted by a liquid on the surface of object, when it is immersed by a fluid is called buoyant force/Upward Thrust.

⇒ Archimede's principle - When a body is immersed partially or fully it experiences a upward force which is equal to the weight of the displaced fluid by the immersed portion of the body.

$$W = mg$$

↓

$$T = W = mg$$

$$\boxed{T = W = PVg}$$

Sound

⇒ Sound - Sound is a wave produced by the vibrating object.

• It is a longitudinal wave.

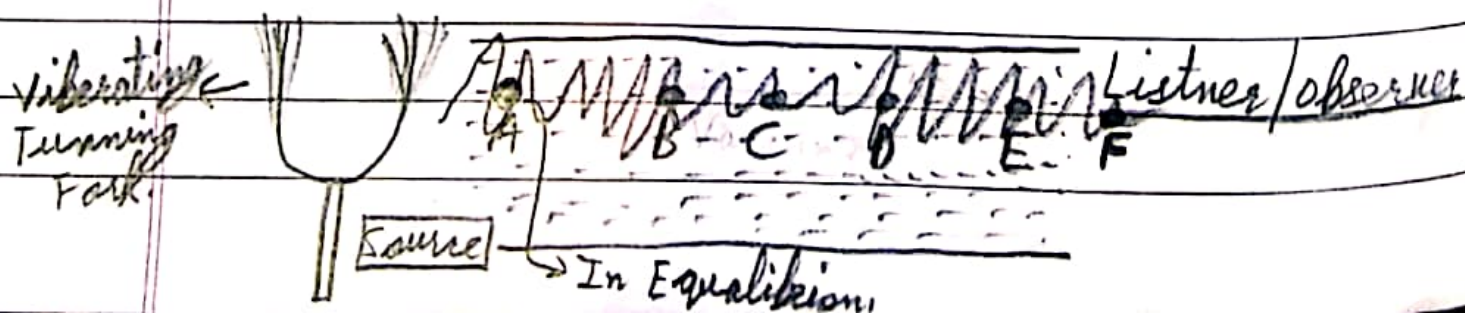
⇒ Production of sound - Sound is produced by the vibrating object.

e.g. Vibrating tuning fork.

⇒ Propagation of sound - The sound needs medium (matter or surrounding) for its propagation i.e. solid, liquid, gas.

⇒ Sound can not propagate through vacuum.

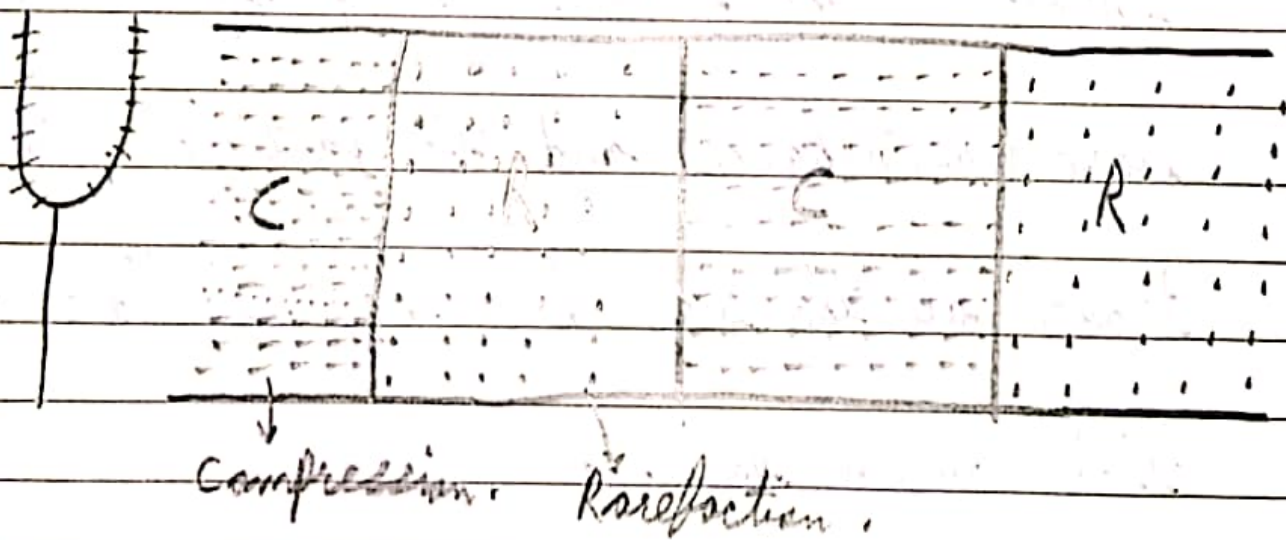
⇒ How it propagates -



- The particles are vibrating with greater Amplitude therefore disturbs the equilibrium of the neighbour particle.

Note:- The particle did not move/travel all the way from source to the ~~last~~ listener but the disturbance travel from source to listener.

⇒ Sound is a longitudinal:



→ Compression: The Region of high pressure in a longitudinal wave is called as compression/

The region where density of particles is high.

→ Rarefaction: The region of low pressure in a longitudinal wave is called rarefaction, OR the region where density of particles of medium is low.

⇒ In Longitudinal wave the particles of medium vibrates in the direction of propagation of wave (disturbance) and the particles did not travel all the way but they oscillate about their mean position.

⇒ Speed of Sound Wave:

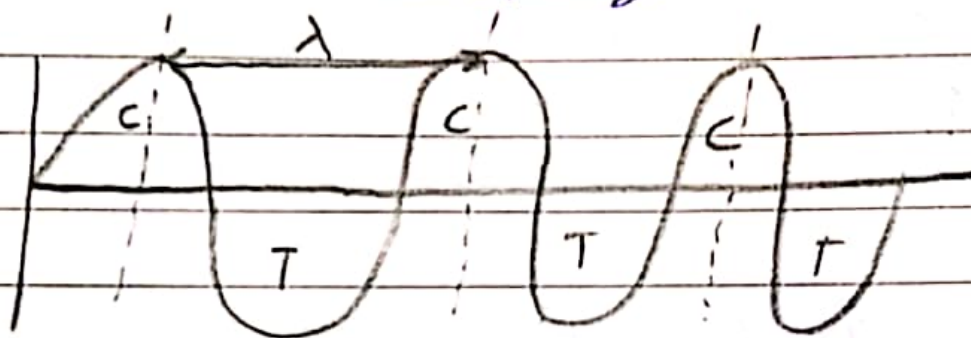
• Speed is defined as distance travelled per

unit time.

$$\Rightarrow v = \frac{\lambda}{T} \rightarrow \text{Wave length.}$$

$T \rightarrow \text{Time}$

$\Rightarrow$ Wave Length: Distance between two successive crests/Troughs.



- Time Period: Time taken to complete one cycle.

$\Rightarrow$ Frequency: Number of cycle per second.
(F).

$$\Rightarrow f = \frac{1}{T} \quad \text{Unit} = s^{-1}$$

(Unit Hertz).

$\therefore$ As we know $v = \frac{\lambda}{T}$

also $\frac{1}{T} = f$

$\boxed{v = \lambda f}$ - Speed = Wave Length $\times$ Frequency.

e.g. A sound wave has a frequency of two kilo Hertz (2 kHz) and wave length 35 cm . How long will it take to travel 1.5 km .

Ans) $f = 2\text{ kHz}$
 $\lambda = 35\text{ cm} = 0.35\text{ m}$
 $T = ?$
 $d = 1.5\text{ km}$

$$\Rightarrow v = \frac{d}{t}, \quad t = \frac{d}{v} \quad \text{--- ①}$$

$$\Rightarrow v = \lambda \times f$$

$$v = 0.35 \times 2000\text{ Hz}$$

$$v = 700\text{ m/s}$$

$\Rightarrow$ Put value of v in ①

$$t = \frac{1500}{700}$$

$$t = 2.14$$

⇒ Speed of sound in different media.

- Speed of sound is different in different media such as:

Speed of Sound in air is 33.1 m/s at 0°C .

- The speed of sound depends upon the properties of medium through which it travels. like, inertia, elasticity, temperature.
- speed of sound is greater in solids as compared in liquid and gases.

$$(V)_{\text{solid}} > (V)_{\text{Liquid}} > (V)_{\text{gas}}$$